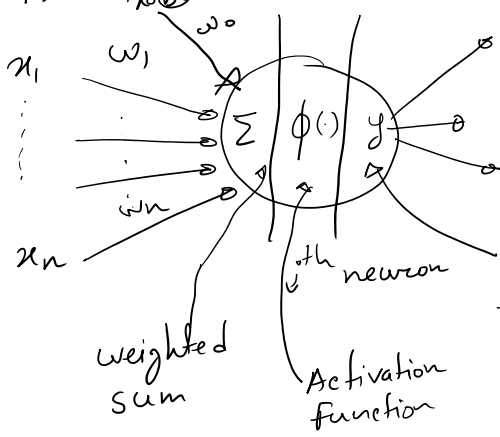


Basic Building Block.

Perceptron

How's it look like?



Directed weighted graph.

$$v_j = \sum_{i=1}^n w_i x_i + x_0 w_0$$

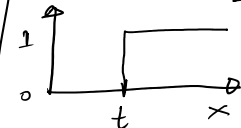
$$y_j = \phi(v_j)$$

$$b_0 = x_0 w_0$$

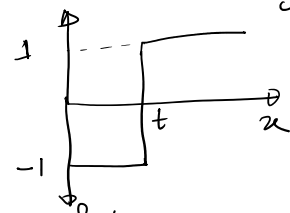
$$v_j = \sum_{i=1}^n w_i x_i + x_0 w_0 = \sum_{i=0}^n w_i x_i$$

$$f(x) = \begin{cases} 1 & x \geq t \\ 0 & x < t \end{cases}$$

Step

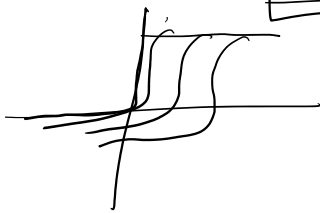


Sign



$$x_t = x_0 e^{-t}$$

non linear



realistic

Sigmoid



$$f(x) = \frac{1}{1 + e^{-ax}}$$

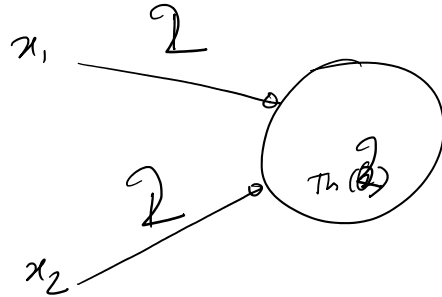
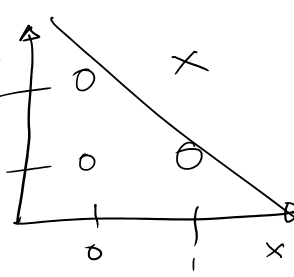
a → ∞ Sigmoid → Step/threshold

Linear Separability

AND

x_1	x_2	x_1, x_2	c_j
0	0	0	0
0	1	0	0
1	0	0	0
1	1	1	1

Feature Map



$$1 \cdot x_1 + 1 \cdot x_2 \geq 2$$

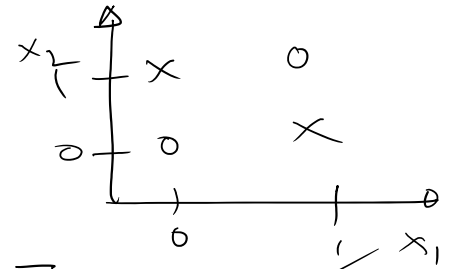
$$2 \geq 2$$

$$b_0 = -2$$

$$2x_1 + 2x_2 \geq 2$$

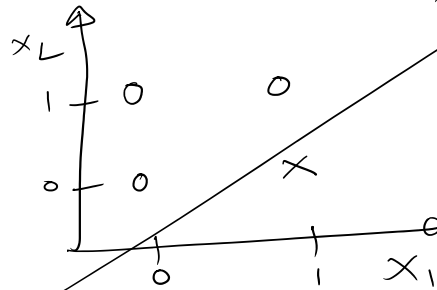
$$x_1 + x_2 \geq 1$$

x_1	x_2	$x_1 \oplus x_2$
0	0	0
0	1	1
1	0	1
1	1	0



x_1 AND NOT $x_2 \rightarrow x_1 \bar{x}_2$

x_1	x_2	$\bar{x}_2$	$x_1 \bar{x}_2$
0	0	1	0
0	1	0	0
1	0	1	1 ✓
1	1	0	0 ✓

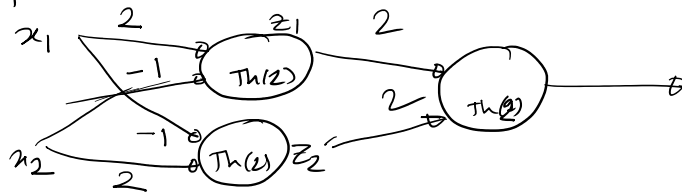


$$2x_1 + (-1)x_2 \geq 2$$

2 -1
①



x_1	x_2	$\bar{x}_1$	$\bar{x}_2$	$x_1 \bar{x}_2$	$\bar{x}_2 \bar{x}_1$	$x_1 \bar{x}_2 + \bar{x}_2 \bar{x}_1$
0	0	1	1	0	0	0
0	1	1	0	0	1	1
1	0	0	1	1	0	1
1	1	0	0	0	0	0



Perceptron

$$v_j = \sum_{i=1}^n w_i x_i \quad \text{weighted sum}$$

$$\hat{y}_j = \phi(v_j) \quad \text{Activation func}$$

$$\hat{y}_j \quad \text{Output func.}$$

Loss function

$$L(y, \hat{y}_j) \quad \text{computed value}$$

truth value given in D.S

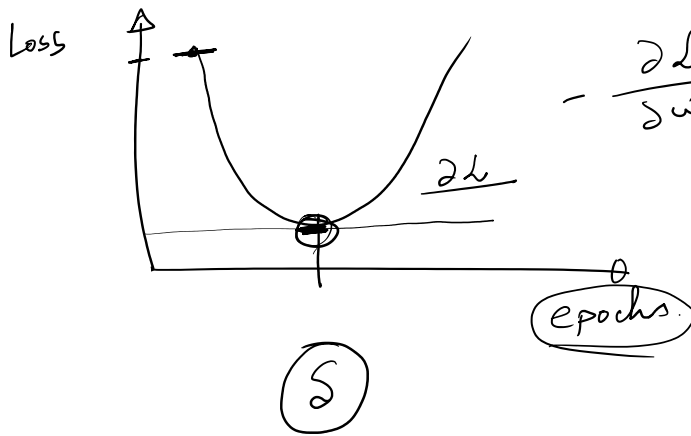
$$= \frac{1}{2} \sum (y - \hat{y}_j)^2$$

$$\frac{\partial L}{\partial w_{ij}}$$

parameters
 ① weights (w)
 ② bias (b)

hyperparameters
 ① D.S ③ epochs
 ② Learning rate (λ)

Problems: Overfitting
 Underfitting.



$$-\frac{\partial L}{\partial w} = 0 \quad \text{④ layer}$$

$$\delta_j = -\frac{\partial E}{\partial v_j}$$

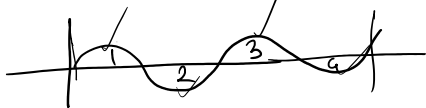
local induced error

local induced Error:

- ① Output layer
- ② Hidden "

Output layer local induced error calculation:-

$$E = \frac{1}{2} \sum_j e_j^2$$



$$\frac{d}{dx}(\sin x) = \cos x$$

$$y_j = \phi(v_j)$$

$$\frac{\partial y_j}{\partial v_j} = \phi'(v_j)$$

$$\frac{\partial E}{\partial e_j} = \frac{\partial}{\partial e_j} \left(\frac{1}{2} (e_1^2 + e_2^2 + \dots + e_n^2) \right)$$

$$= \frac{1}{2} \frac{\partial}{\partial e_j} e_j^2$$

$$= \frac{1}{2} \cdot 2 \cdot e_j = e_j$$

$$\frac{\partial e_j}{\partial \phi(v_j)}$$

$$\delta_j = -\frac{\partial E}{\partial v_j}$$

$$= -\frac{\partial E}{\partial e_j} \frac{\partial e_j}{\partial y_j} \frac{\partial y_j}{\partial v_j}$$

$$= -\frac{\partial E}{\partial e_j} (-1) \phi'(v_j)$$

$$\frac{\partial y_j}{\partial v_j} = \phi'(v_j)$$

$$= -\frac{\partial}{\partial e_j} (-1) \phi'(v_j)$$

$$\delta_j = (d_j - y_j) \phi'(v_j) = \left[\frac{\partial E}{\partial e_j} \right] \phi'(v_j)$$

$$= (d_j - y_j) \phi'(v_j) = e_j \phi'(v_j)$$

$$\phi(\cdot) = \text{sigmoid}$$

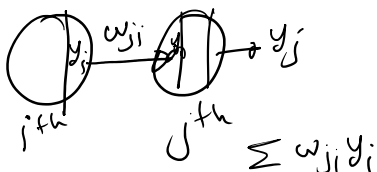
$$\phi(x) = \frac{1}{1 + e^{-ax}}$$

$$\begin{aligned} \frac{d\phi(x)}{dx} = \phi'(x) &= \frac{d}{dx} \left(\frac{1}{1 + e^{-ax}} \right) \\ &= \frac{d}{dx} (1 + e^{-ax})^{-1} = (-1) (1 + e^{-ax})^{-2} (-a) e^{-ax} \\ &= a \frac{e^{-ax}}{(1 + e^{-ax})^2} \\ &= a \frac{(1 + e^{-ax}) - 1}{(1 + e^{-ax})(1 + e^{-ax})} \end{aligned}$$

$$\begin{aligned} &= a \left[\frac{1 + e^{-ax}}{(1 + e^{-ax})^2} - \frac{1}{(1 + e^{-ax})^2} \right] \\ &= a \left[\frac{1}{1 + e^{-ax}} \left[1 - \frac{1}{1 + e^{-ax}} \right] \right] \\ &= a \left[\phi(x) [1 - \phi(x)] \right] \end{aligned}$$

local induced error.

$$\delta_j = \begin{cases} (d_j - y_j) \phi'(v_j) & \text{output layer} \\ ? & \text{hidden layer} \end{cases}$$



$$\begin{aligned} \Delta w_{ji} &= -\eta \frac{\partial E}{\partial w_{ji}} = -\eta \frac{\partial E}{\partial y_j} \frac{\partial y_j}{\partial w_{ji}} \\ w_{\text{new}} &= w_{\text{old}} + \Delta w_{ji} = \eta \delta_j (y_i) \end{aligned}$$

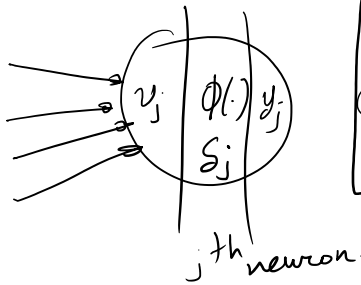
$$\Rightarrow w_{\text{new}} = w_{\text{old}} + \eta \delta_j y_i$$

Hebbs Learning.

Local induced Error

Sunday, June 28, 2026 7:54 AM

Local induced error



jth neuron
output layer

$$\begin{aligned} ① v_j &= \sum w_{ji} y_i \\ ② y_j &= \phi(v_j) \end{aligned}$$

$$\delta_j = - \frac{\partial E}{\partial v_j}$$

$$= - \left[\frac{\partial E}{\partial e_j} \right] \left[\frac{\partial e_j}{\partial y_j} \right] \left[\frac{\partial y_j}{\partial v_j} \right]$$

$$③ E = \frac{1}{2} \sum_{j=1}^n e_j^2 = - e_j (-1) \phi'(v_j)$$

$$④ e_j = d_j - y_j \rightarrow \delta = e_j \phi'(v_j)$$

① nonlinear - sigmoid

$$\phi'(v_j) = \phi(v_j) (1 - \phi(v_j))$$

$$\frac{\partial y_j}{\partial v_j} = \phi'(v_j)$$

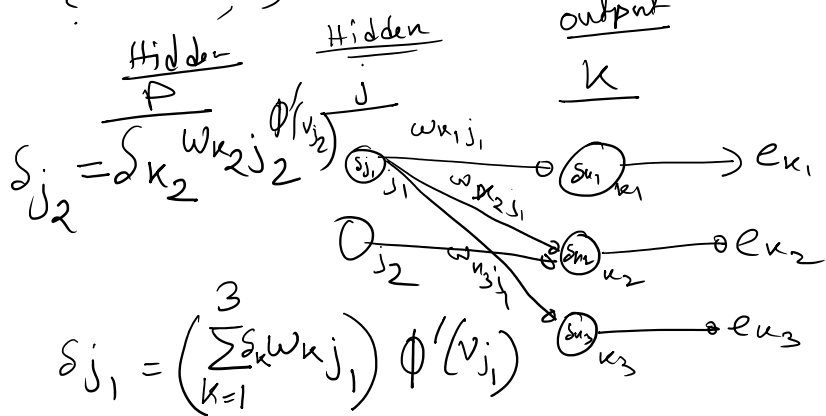
$$\frac{\partial e_j}{\partial y_j} = 0 - 1 = -1$$

$$E = \frac{1}{2} (e_1^2 + e_2^2 + \dots + e_j^2 + \dots + e_n^2)$$

$$\frac{\partial E}{\partial e_j} = \frac{1}{2} \cdot 2 \cdot e_j = e_j$$

$$\delta_j = \begin{cases} e_j \phi'(v_j) & ; \text{jth neuron is in output layer} \\ ? & ; \text{jth neuron is in hidden neuron} \end{cases}$$

Solu:



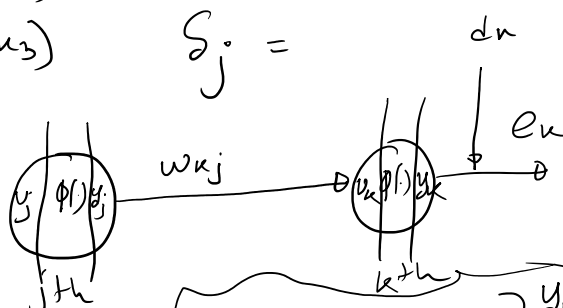
$$\delta_{j_1} = ?$$

$$\begin{aligned} \delta_{u_1} &= e_{u_1} \phi'(v_{u_1}) \\ \delta_{u_2} &= e_{u_2} \phi'(v_{u_2}) \\ \delta_{u_3} &= e_{u_3} \phi'(v_{u_3}) \end{aligned}$$

$$\delta_{j_1} = \left(\sum_{k=1}^3 \delta_{u_k} w_{kj_1} \right) \phi'(v_{j_1})$$

$w_{To From}$

$$\delta_j =$$



$$\begin{aligned} v_k &= \sum w_{kj} y_j \\ y_k &= \phi(v_k) \end{aligned}$$

$$\delta_j = - \frac{\partial E}{\partial v_j} = \left[\frac{\partial E}{\partial e_k} \cdot \frac{\partial e_k}{\partial y_k} \cdot \frac{\partial y_k}{\partial v_k} \right] \left[\frac{\partial v_k}{\partial y_j} \right] \left[\frac{\partial y_j}{\partial v_j} \right]$$

$$\delta_j = - \frac{\partial E}{\partial v_j} = \underbrace{\left(\frac{\partial L}{\partial e_k} \cdot \frac{\partial e_k}{\partial y_k} \cdot \frac{\partial y_k}{\partial v_j} \right)}_{\delta_k} \underbrace{\left(\frac{\partial y_j}{\partial v_j} \right)}_{\sum_k \omega_{kj} \phi'(v_j)}$$

$$= \left(\sum_k \delta_k \omega_{kj} \right) \phi'(v_j)$$

$$\delta_j = \begin{cases} e_j \phi'(v_j) \\ \boxed{\sum_k \omega_{kj} \phi'(v_j)} \end{cases} ; j^{\text{th}} \text{ neuron is in hidden layer}$$

$$w_{\text{new}} = w_{\text{old}} + \Delta w_{ji}$$

$$\Delta w_{ji} = -\eta \frac{\partial E}{\partial w_{ji}} = \eta \left(-\frac{\partial E}{\partial v_j} \right) \cdot \frac{\partial v_j}{\partial w_{ji}}$$

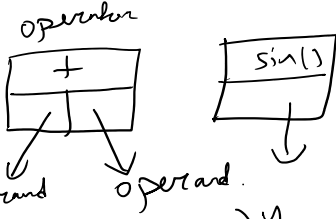
$$w_{\text{new}} = w_{\text{old}} + \eta \delta_j y_i$$

Computational Graph (CG)

Operator

$+$, $*$, $/$, $-$ binary

$\sin$, $\cos$, $\exp$ unary



Operand

x, y, z independent

t_1, t_2, t_3 dependent



operator

Forward pass $\frac{\partial F}{\partial x} = 0 + (-\sin(xy))y$

Reverse $\frac{\partial L}{\partial x} = 1$

$f(x, y, z) = \sin(yz) + \cos(xy) + e^{x+y}$

MIPS

$t_1 = y * z$

$t_2 = x * y$

$t_3 = x + y$

$t_4 = \sin(t_1)$

$t_5 = \cos(t_2)$

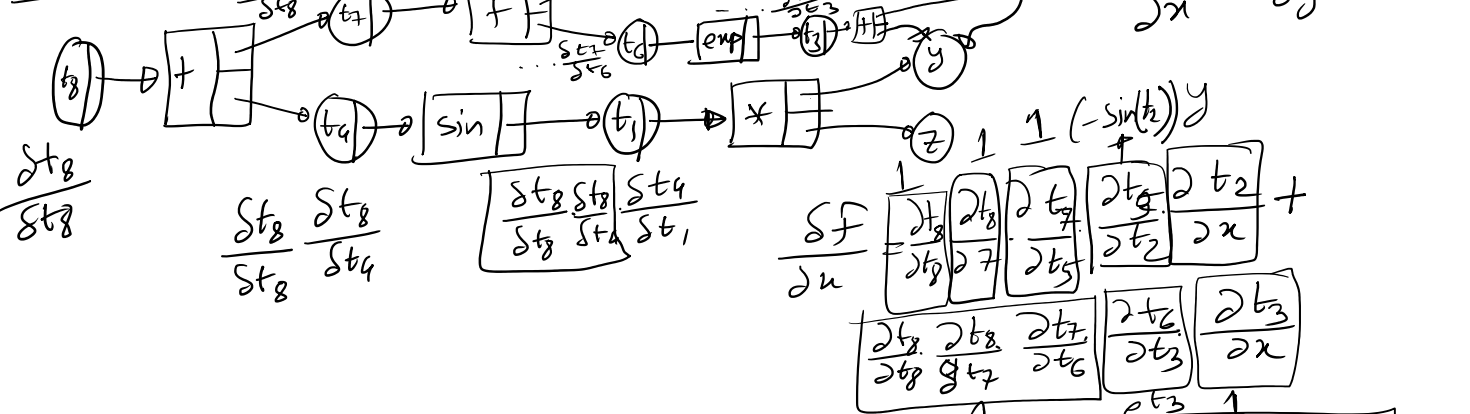
$t_6 = \exp(t_3)$

$t_7 = t_5 + t_6$

$t_8 = t_7 + t_4$

$\frac{\partial t_8}{\partial x} = \frac{\partial t_7}{\partial x} + \frac{\partial t_4}{\partial x}$

Reverse



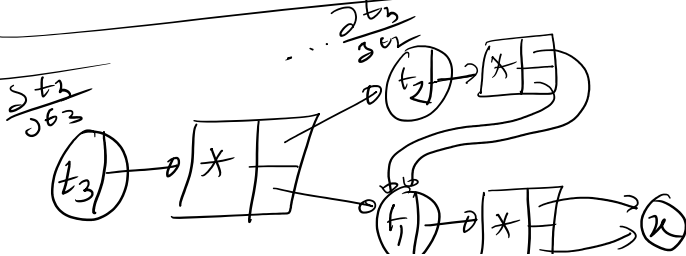
$= -y \sin(xy) + e^{x+y}$

$f(x) = x^6$

$\frac{\partial F}{\partial x} = 6x^5$

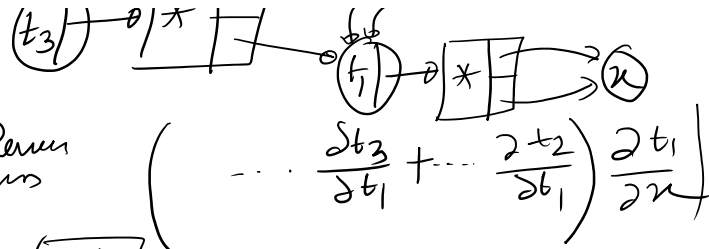
$t_1 = x * x$

$t_2 = x * t_1$



$\frac{\partial L}{\partial u}$

$$\begin{aligned} t_1 &= x * x \quad // x^2 \\ t_2 &= t_1 * t_1 \quad // x^4 \\ t_3 &= t_1 * t_2 \quad // x^6 \end{aligned}$$



$$\begin{aligned} \frac{\partial t_3}{\partial x} &= \left(\frac{\partial t_3}{\partial t_2} \frac{\partial t_2}{\partial t_1} + \frac{\partial t_3}{\partial t_1} \right) \left(\frac{\partial t_1}{\partial x} \right) \\ &= (1 \cdot t_2 + 1 \cdot t_1 \cdot 2t_1) (2x) \\ &= (x^4 + 2x^6) 2x \\ &= 2x^5 + 4x^7 \\ &= 6x^5 \end{aligned}$$

$5x^4$

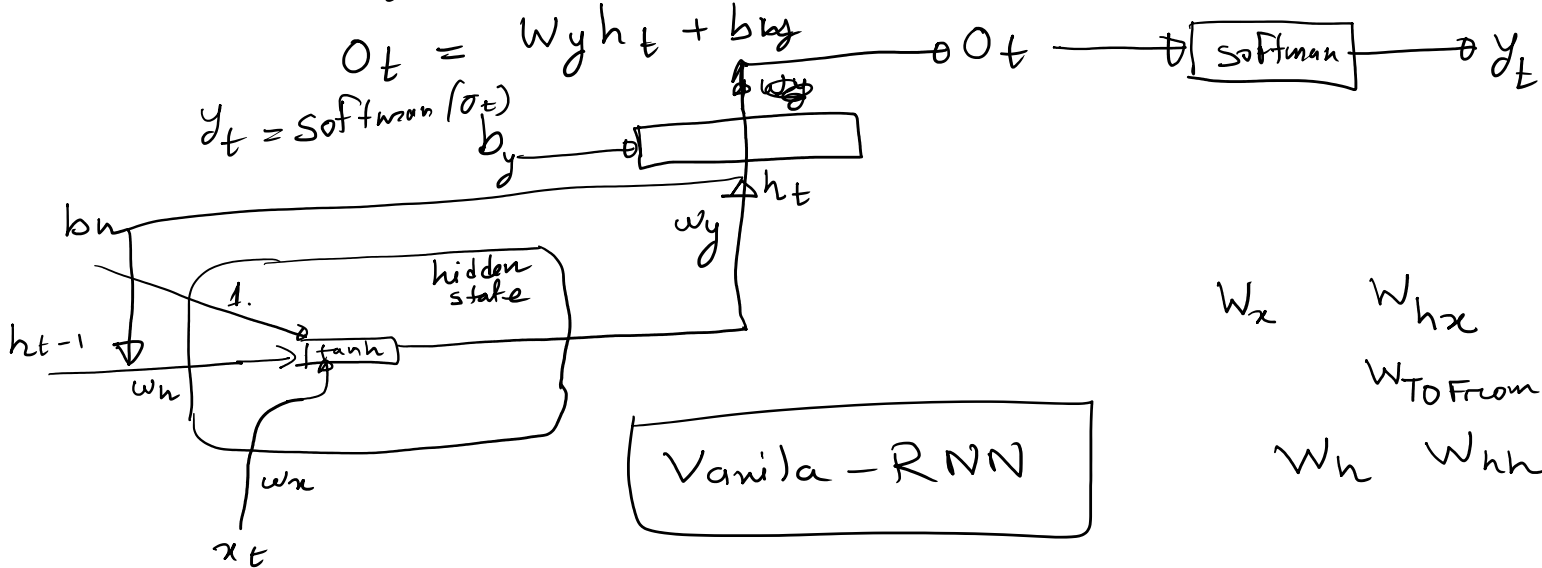
$$\begin{aligned} t_1 &= x * x \\ t_2 &= t_1 * t_1 \\ t_3 &= t_2 * x \end{aligned}$$

- Perceptron
- MLP
- CG
- Process Training
- Optimization
- Regularization
- Dropout
- RNN
- LSTM
- GRU
- Midterm

$$h_t = \tanh(w_x x_t + w_h h_{t-1} + b_h)$$

$$o_t = w_y h_t + b_y$$

$$y_t = \text{softmax}(o_t)$$



w_x w_{hx}
 w_{ToFrom}
 w_h w_{hh}